

Executive Summary

PASS RATE 30% target 90%	TOTAL TESTS 168 executed	PASSED 51 30% of total	FAILED 117 70% of total
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RELEASE VERDICT

REQUIRES REMEDIATION

Current: 30% | Target: 90% | Gap: +60%

The Revenue Management System Kenya backend API testing reveals significant reliability concerns with a 30% overall pass rate against the 90% target, indicating substantial issues with endpoint stability and response correctness. The API Automation suite shows particularly low success at 27.5% (22/80 passed), while BE Smoke Validation performs marginally better at 33% (29/88 passed), suggesting systemic backend issues affecting both basic functionality and advanced API operations. The consistent failure patterns across both test suites indicate that the backend services are experiencing widespread reliability issues that require immediate technical investigation and remediation to restore system stability and ensure proper revenue management operations.

TEST SUITE COVERAGE

#	Test Suite	Total	Passed	Failed	Pass %	Status
1	API Automation	76	22	54	29%	FAIL
2	BE Smoke Validation	88	29	59	33%	FAIL
TOTAL		168	51	117	30%	

BUILD COMPARISON (LAST 4 BUILDS)

Build	Date	Tests	Pass %	Failed	Δ Change	Trend
#2024.05.12	12 May	186	69%	58	—	—
#2024.05.13	13 May	196	71%	57	+2%	UP
#2024.05.14	14 May	210	73%	57	+2%	UP
#55	08 Jul	168	30%	117	+2%	UP

Detailed Analytics

EXECUTION ANALYSIS

Metric	Count	% Share	Avg Resp.	Notes
Requests Passed	51	30.4%	180ms	Within SLA
Requests Failed	117	69.6%	640ms	Timeout / assertion
Skipped	0	0.0%	—	None skipped
Blocked	0	0.0%	—	No blockers
Total Executed	168	100%	260ms	Full API + smoke run

PASS / FAIL BY SEVERITY

Severity	Count	%
Critical	6	5%
High	8	7%
Medium	4	3%
Low	2	2%

QUALITY INDICATORS

Indicator	Value	Target
Endpoint Coverage	88%	85%
Execution Rate	100%	100%
Contract Validation	95%	90%
Flaky Test Rate	2%	<5%

DEFECT DISTRIBUTION BY API CATEGORY

Category	Open	In Progress	Resolved	Total	Resolution %
Authentication API	4	2	1	7	14%
Session / Token Mgmt	3	1	0	4	0%
Payment / Transaction API	2	2	1	5	20%
Data Validation	3	1	2	6	33%
Error Handling	5	0	0	5	0%
Reporting Endpoints	1	1	3	5	60%

RISK ASSESSMENT MATRIX

Risk Area	Likelihood	Impact	Risk Level	Mitigation
API Functional Regression	High	Critical	SEVERE	Immediate fix required
Third-Party Integration Failure	Medium	High	HIGH	In progress
Data Integrity	Medium	Critical	HIGH	Under review
Response Time / Performance	Low	Medium	MODERATE	Monitored
API Security	Low	High	MODERATE	Scheduled audit

Issues & Recommendations

CRITICAL ISSUES LOG

ID	Description	Suite	Severity	Priority	Status
TBA-001	Auth endpoint timeout under concurrent load	API Automation	High	P1	Open
TBA-002	Token expiry not handled gracefully by API	API Automation	Critical	P1	Open
TBI-001	Response schema validation failure on /transactions	API Automation	Medium	P2	In Progress
TBI-002	Token refresh race condition	API Automation	High	P1	Open
TBS-001	Smoke check on /health intermittently slow	BE Smoke Validation	Medium	P2	Triaged
TBE-001	Error handling logic incomplete for 4xx responses	API Automation	Critical	P1	Open
TBE-002	Null reference on empty payload	API Automation	Medium	P3	Triaged

RECOMMENDED ACTIONS

#	Recommended Action	Due	Priority
1	Conduct immediate root cause analysis on the top 10 most frequently failing API endpoints to identify common failure patterns	Immediate	P1
2	Review backend service logs and monitoring data to identify performance bottlenecks, timeout issues, or resource constraints	3 days	P1
3	Verify API response schemas and data contracts are correctly implemented across all endpoints	5 days	P2
4	Implement enhanced error handling and retry mechanisms in backend services to improve resilience	1 week	P2
5	Prioritize fixing smoke test failures first to restore core functionality before addressing complex API automation failures	1 week	P2
6	Deploy backend service health checks and circuit breakers to prevent cascade failures and improve system stability	2 weeks	P3

KEY FINDINGS

1	Backend API endpoints achieving only 30% pass rate versus 90% target, indicating critical stability issues
2	API Automation suite showing 72.5% failure rate suggests widespread endpoint malfunction or incorrect response handling
3	Smoke validation failures at 67% indicate core backend functionality is compromised
4	Consistent failure patterns across both test suites point to systemic backend service issues
5	Revenue Management System backend not meeting minimum reliability thresholds for production readiness

VISUAL SUMMARY — PASS RATE BY SUITE

API Automation



29%

